Evaluation & Error Analysis – Week 5

Group	Group 2 - AI Exam Answer Grader
Roll Numbers	J008, J009, J010, J011, J012
Class & Course	B.Tech DS 4th year – Speech & Natural Language Processing (SNLP)

1. Objective and Test Set

This report evaluates the Week 4 hybrid NLP grading model (V2: TF-IDF + sentence-embedding similarity features, 7 features per answer) on the held-out EngSAF test set of 506 student answers. Each answer is graded 0 = Incorrect, 1 = Partially correct or 2 = Correct against a reference answer.

Class	Label	Count	Share of test set
Incorrect	0	113	22.33%
Partially correct	1	191	37.75%
Correct	2	202	39.92%
Total		**506**	**100%**

2. Quantitative Evaluation

Macro-F1 weights the three classes equally; QWK (quadratic weighted kappa) measures agreement with the human grade and penalises large errors more; MAE is the mean error in label points (lower is better); partial recall is the share of partially correct answers recognised. The values below are identical to the V2 test results reported in Week 4.

Metric	Value	What it means
Accuracy	0.5257	266 of 506 answers graded exactly right
Macro-F1	0.5005	F1 averaged equally over the 3 classes
QWK	0.3643	Agreement with human grades (distant errors penalised more) — only fair
MAE	0.5711	Average distance of 0.57 grade levels from the human grade
Partial Recall	0.3770	Only 72 of 191 partially correct answers recognised

Per-class F1 (from the confusion matrix): Incorrect 0.447, Partial 0.430, Correct 0.625. The model beats a majority-class guess (39.9%) but is not reliable enough for unsupervised grading; the partially correct class is the weakest.

3. Confusion Matrix

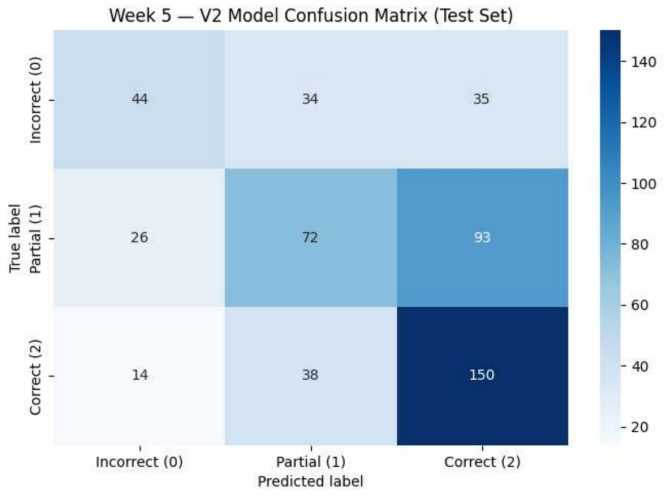


Figure 1 – Heat map of the V2 confusion matrix on the test set.

- **Over-grading dominates:** 162 errors grade too high, against 78 too low.
- **Partial is the main confusion:** 93 of 191 (48.7%) partially correct answers are predicted Correct.
- **49 two-level errors** (35 Incorrect → Correct, 14 Correct → Incorrect) push MAE above the error rate.

- **Incorrect answers are poorly recognised:** only 44 of 113 (38.9%) are found, and 35 are graded Correct.

4. Error Analysis

The model made **240 errors out of 506 (47.43%)**. The table splits them by the direction of the mistake.

Error type (true → predicted)	Count	% of errors
Partial → Correct	93	38.75%
Correct → Partial	38	15.83%
Incorrect → Correct	35	14.58%
Incorrect → Partial	34	14.17%
Partial → Incorrect	26	10.83%
Correct → Incorrect	14	5.83%

Confidence: correct predictions average 0.585 (median 0.567) versus 0.518 (median 0.498) for errors. Confidence carries some signal, but the ranges overlap (errors reach 0.884), so it cannot reliably flag wrong grades on its own.

5. Robustness

- **Incomplete answers** are promoted to Correct – the model measure similarity, not coverage of key points.
- **On-topic but wrong answers** earn credit because they reuse the reference vocabulary.
- **Negation and contradiction** are missed: answers saying "True" where the reference says "FALSE" can still be graded Correct.
- **Concise or paraphrased correct answers** are under-graded (38 cases), so wording variation matters.
- **Failures are spread across topics** (operating systems, hydrology, computer vision, Industry 4.0), so the cause is the similarity-based features rather than one question.

6. Failure Cases

Ten errors from the test set, one for each of 10 different topics (the highest-confidence genuine error per topic, selected in the notebook): 4 Incorrect → Correct, 4 Correct → Partial and 2 Partial → Correct. T = true score, P = predicted score; answer text is truncated as in the notebook output.

#	True → Predicted	Topic / question	Student answer (start)	T	P	Conf.	Why it failed
1	Partial → Correct	Spinlock (Q1 T/F)	True, A spinlock is called spinlock because we...	1	2	0.73	Shared terminology taken as complete; verdict "Tr vs reference "FALSE" not detected
2	Incorrect → Correct	ISA (Q1 T/F)	False. Abstractions are provided by the hardwa...	0	2	0.58	On-topic hardware wording accepted; two-level er
3	Incorrect → Correct	Canal fall situation	when there is a chance for water in the canal ...	0	2	0.46	Topically close but does not match the expected condition (natural slope); wrong even at low confidence
4	Incorrect → Correct	Role of edge direction in Harris corner detection	In the Harris corner Detector Algorithm, there...	0	2	0.72	Correct domain vocabulary taken as a correct explanation
5	Correct → Partial	Two advantages of Industry 4.0	Faster manufacturing, less wastage	2	1	0.47	Short answer, low overlap with reference wording
6	Incorrect → Correct	Rating curve validity	The Rating curve will not be correct if its ex...	0	2	0.68	Similar wording to reference despite wrong conten high-confidence two-level error
7	Correct → Partial	Coagulation and alkalinity	le chateliers principle so more hydrolysis inc...	2	1	0.39	Correct idea phrased very differently from the reference; low confidence
8	Correct → Partial	Vapour pressure deficit	difference between saturate and actual vapor p...	2	1	0.38	Valid definition worded differently from the refere
9	Correct → Partial	Factors affecting evaporation	surface area of body, depth of body, temperature...	2	1	0.45	Valid factors listed, but different from the referenc list, so overlap is low
10	Partial → Correct	Muskingum flood routing assumptions	Assumption of Muskingum flood routing method. ...	1	2	0.53	Restates the topic and overlaps with the reference, is incomplete

7. Conclusion

The V2 model reaches 52.57% accuracy, Macro-F1 0.5005 and QWK 0.3643 on 506 test answers, with 37.70% recall on partially correct answers. Its main weakness is promoting partial or on-topic wrong answers to Correct (93 + 35 errors) because the features measure similarity rather than correctness and completeness. Next steps: key-point coverage checks against the mark scheme, an entailment/contradiction feature for negation, question-specific calibration, ordinal or class-weighted training for the partial class, and human review of low-confidence or borderline grades.

#Additional Note:

We are extremely sorry sir that we couldn't implement your feedback into our current code as we have been a bit busy with placements and mid terms exam but we have taken your feedback into consideration and are actively researching upon the same to ensure we get better accuracies for our project. We request you to grant us some more time so that we can work upon on this.